RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 9A: Integration Techniques

1. Integrate the following with respect to x.

(a) $\left(1-\frac{1}{x}\right)^2$ **(b)** $x\sin 3x$ **(c)** $\frac{x^2}{\mathrm{e}^x}$ **(d)** $\frac{x}{1+x}$

(e) $3\ln x^2$ **(f)** $\sin x\cos^4 x$ **(g)** $\frac{x}{9x^2+1}$ **(h)** $\frac{1}{(4-5x)}$

(i) $\frac{1}{(4-5x)^2}$ **(j)** $\sin^2 2x$ **(k)** $\frac{x}{x^4+4}$ **(l)** $\frac{x}{x^4-4}$

(m) $\tan 2x\sec 2x$ **(n)** $\frac{(x+1)^2}{x^2+1}$ **(o)** $\cot^2 3x$ **(p)** $\left(\mathrm{e}^x+x\right)^2$

(q) $\frac{1}{x^2(x-1)}$ **(r)** $\frac{1}{\sqrt{1-2x^2}}$ **(s)** $\frac{x}{\sqrt{1-2x^2}}$ **(t)** $x\cos(\frac{x^2+1}{2})$

(u) $\frac{3x^2+x+4}{x^3+x}$

2. By expressing as $R\cos(\theta+\alpha)$, where R and α are constants to be determined, find $\int\frac{1}{\cos\theta-\sqrt{3}\sin\theta}\mathrm{d}\theta$. [2]

3. **(a)** $\int x^2\cos 2x\,\mathrm{d}x$ **(b)** $\int x\sqrt{x^2+1}\,\mathrm{d}x$ **(c)** $\int x\sin x^2\,\mathrm{d}x$

(d) $\int\frac{2}{\sqrt{9-4x^2}}\,\mathrm{d}x$ **(e)** $\int\frac{x}{(x+1)^2}\,\mathrm{d}x$ **(f)** $\int x^3\mathrm{e}^{2x}\,\mathrm{d}x$

4. By using the identity $\sin^2 x+\cos^2 x=1$, determine $\int\sin^3 x\,\mathrm{d}x$.

5. By using the given substitution, determine the following integrals, leaving your answers in the exact form whenever applicable.

(a) $\int\tan\frac{x}{2}\sec^4\frac{x}{2}\mathrm{d}x \quad [u=\sec\frac{x}{2}]$	**(b)** $\int_1^{\mathrm{e}}\frac{1}{y\sqrt{1-(\ln y)^2}}\mathrm{d}y \quad [u=\ln y]$
(c) $\int_0^1\frac{1}{(4-t^2)^{\frac{3}{2}}}\mathrm{d}t \quad [t=2\sin\theta]$	**(d)** $\int_2^3\sqrt{\frac{x-2}{4-x}}\,\mathrm{d}x \quad [x=2+2\cos^2\theta]$

Note: Try solving **(a)** and **(b)** without using substitution.

6. Determine the exact value of $\int_0^1 \frac{x+9}{2x^2+x-6}\,\mathrm{d}x$. [3]

7. By the use of the substitution $x = 3\sin t$, determine the exact value of $\int_0^3 x^2\sqrt{9-x^2}\,\mathrm{d}x$. [4]

8. Given that $\int_0^a x\sin x\,\mathrm{d}x = 0.5$, where $0 < a < 2$, find an equation that is satisfied by a and use it to find the value of a. [3]

9. Express $\frac{x^2+x+28}{(1-x)(x^2+9)}$ in partial fractions.

Hence, show that $\int_0^3 \frac{x^2+x+28}{(1-x)(x^2+9)}\mathrm{d}x = \frac{\pi}{12} - 2\ln 2$. [4]

10. By using the substitution $u = \sqrt{x+2}$, or otherwise, show that $\int_2^7 \frac{1}{(x+1)\sqrt{x+2}}\,\mathrm{d}x = \ln\frac{3}{2}$.

Hence, find the exact value of $\int_2^7 \frac{(x-1)}{(x+1)\sqrt{x+2}}\mathrm{d}x$. [5]

11.

(a) $\int \frac{x+2}{\sqrt{1-3x^2}}\mathrm{d}x$ **(b)** $\int \frac{2}{x\ln x^2}\mathrm{d}x$ **(c)** $\int \frac{1}{1+\mathrm{e}^x}\mathrm{d}x$

(d) $\int \frac{1}{x\left[4+(\ln x)^2\right]}\mathrm{d}x$ **(e)** $\int \ln(2x+1)\,\mathrm{d}x$ **(f)** $\int \frac{6x+4}{(1-2x)(1+3x^2)}\mathrm{d}x$

12. By using the substitution $u = x^4+1$, find $\int_1^3 \frac{1}{x(x^4+1)}\mathrm{d}x$, leaving your answers in the exact form. [4]

13. You are given that $I = \int x(1-x)^{\frac{1}{2}}\,\mathrm{d}x$.

(i) Use integration by parts to find an expression for I. [2]

(ii) Use the substitution $u^2 = 1-x$ to find another expression for I. [2]

(iii) Show algebraically that your answers to part **(i)** and **(ii)** differ by a constant. [2]